Workflow Diagrams

Workflow Diagrams

- Course Registration (Frame 1) & Academic Advising (Frame 2)

Course Registration Workflow

The Course Registration Workflow models how students, the course registration system, advisors, and IT staff coordinate to search for courses, validate enrollment requirements, and complete registration. This workflow is important because course registration involves several checks, including schedule conflicts, prerequisite verification, seat availability, and potential advisor overrides. The process organizes these steps into a structured system that automates validations while still allowing administrative intervention when needed.

The process begins when a student logs into the registration system and searches for available courses. The system verifies the student's identity and returns course results along with current seat availability. After reviewing course details, the student adds a course to their schedule cart. The system then automatically checks for schedule conflicts and prerequisite requirements. If a conflict exists, the student must adjust their schedule before continuing. If prerequisites are not met, the student may submit an override request, which is reviewed by an advisor who will send the request to a departmental advisor for approval. If the override is denied, the student is notified.

Once requirements are satisfied, the student submits the registration. The system checks real-time seat availability and either enrolls the student and updates the SIS/LMS systems or places the student on a waitlist if no seats are available. In parallel, the IT department monitors system performance during registration periods. If peak load conditions occur, system resources are automatically scaled to maintain performance, and all actions are logged for auditing and compliance.

Related Backlog Items

Epics:

- T1I3W2-2: Reliable and Fair Course Registration
- T1I3W2-3: Student Planning and Visibility

User Stories:

- T1I3W2-5: Real-time Data

- T1I3W2-10: Prerequisites Checked Automatically
- T1I3W2-11: Granting Overrides
- T1I3W2-43: Registration Confirmation
- T1I3W2-46: Accurate Course Information
- T1I3W2-50: User Authentication

Academic Advising Workflow

The Academic Advising Workflow models how advisors, students, the course registration system, and department administrators coordinate to review a student's academic status and address any issues that could affect their progress or course registration. This workflow is important because advising often requires pulling student information from multiple systems, identifying risks, coordinating appointments, and sometimes requesting administrative overrides. The process diagram organizes these steps into a single structured workflow and clarifies the responsibilities of each participant.

The process begins when an advisor logs into the advising dashboard and searches for a student. The system authenticates the advisor and retrieves the student's degree audit and any flagged academic risks. The advisor reviews the student's information to determine whether the student is high risk or requires additional attention. If a risk is identified, the system sends an automated message prompting the student to schedule an advising appointment. The student receives the message and schedules the meeting, and the advisor confirms the appointment and reviews potential academic issues. If the advisor determines that a registration override is needed, a request is submitted to a department administrator for approval. The administrator reviews the request and either approves or denies it. Finally, the advisor checks for financial or visa compliance issues. If any are present, the student is referred to the appropriate office. Otherwise, the process ends with the student confirmed to be on track academically.

Related Backlog Items

Epics:
- T1I3W2-3: Student Planning and Visibility
- T1I3W2-4: Administrative Control and Oversight

User Stories:
- T1I3W2-11: Granting Overrides
- T1I3W2-22: Automated Enrollment Monitoring
- T1I3W2-24: Financial Status Verification
- T1I3W2-25: Financial Implication Visibility
- T1I3W2-35: Student Academic Progress Visibility

- T1I3W2-36: High-Risk Student Identification
- T1I3W2-45: Student Course Planning

User Interfaces

Student Planning Interface

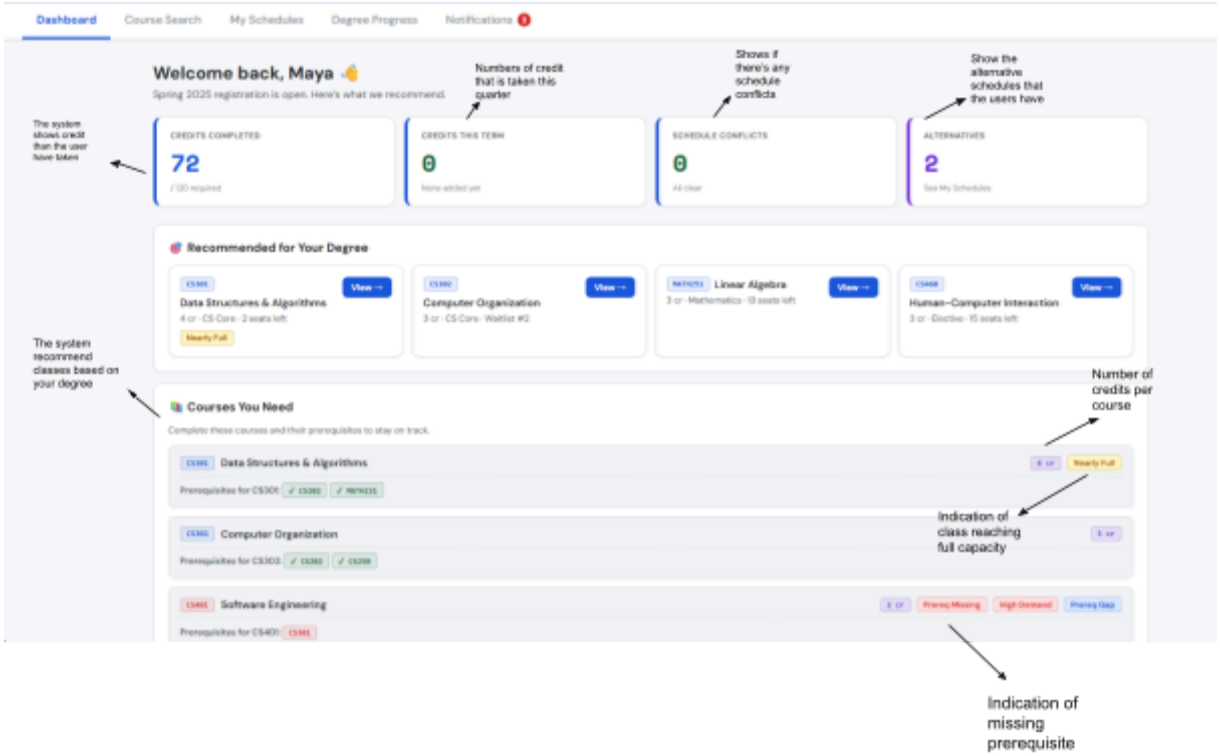
GSU Course Registration System

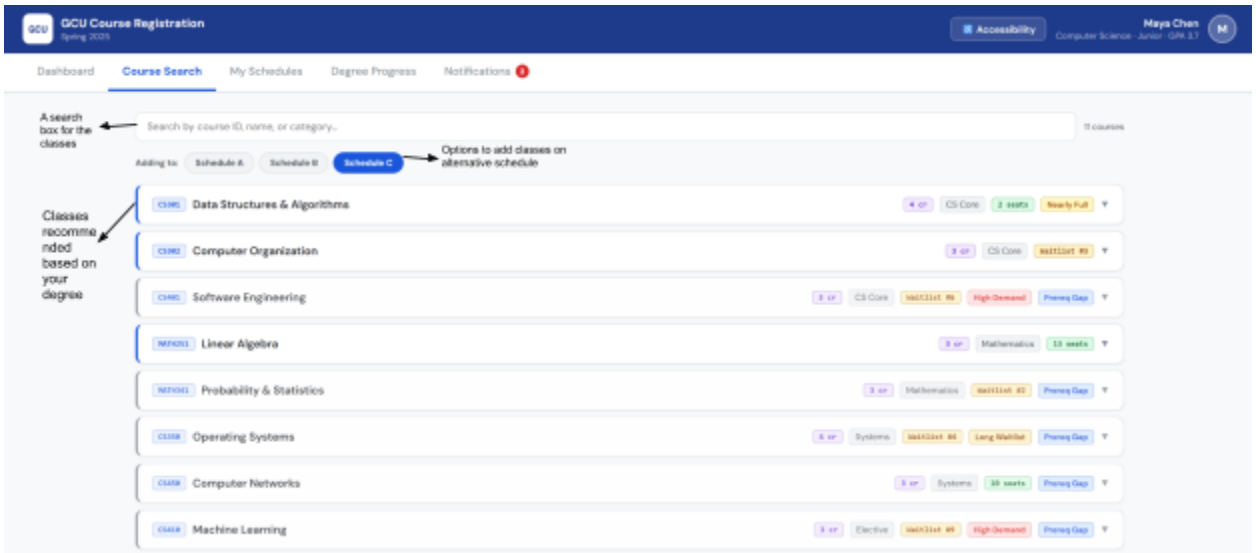
The following prototype illustrates the different interfaces that support student use of the GSU Course Registration System. It shows how students can search for and filter classes, check course prerequisite and enrollment information, plan alternate schedules, track their degree completion progress, and manage notifications.

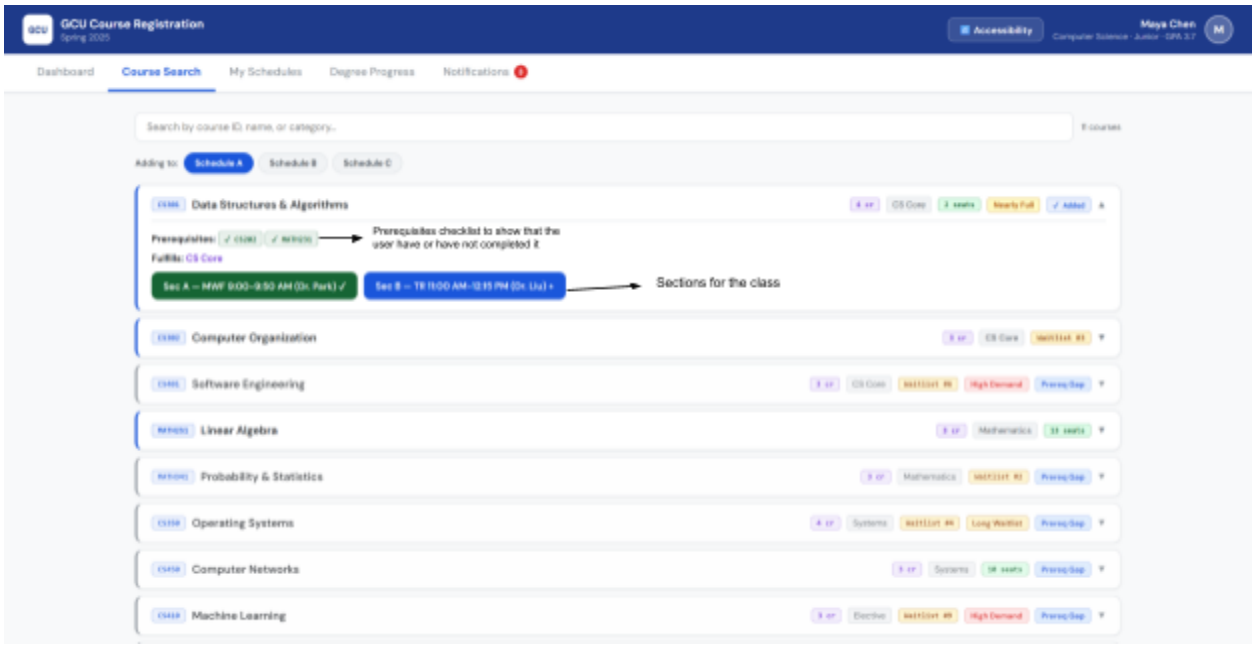
The design prioritizes displaying in-depth, accurate information for every course. The stakeholder interviews identified that this was the primary pain point, where students needed to consult secondary sources or navigate through confusing interfaces in order to locate valuable information.

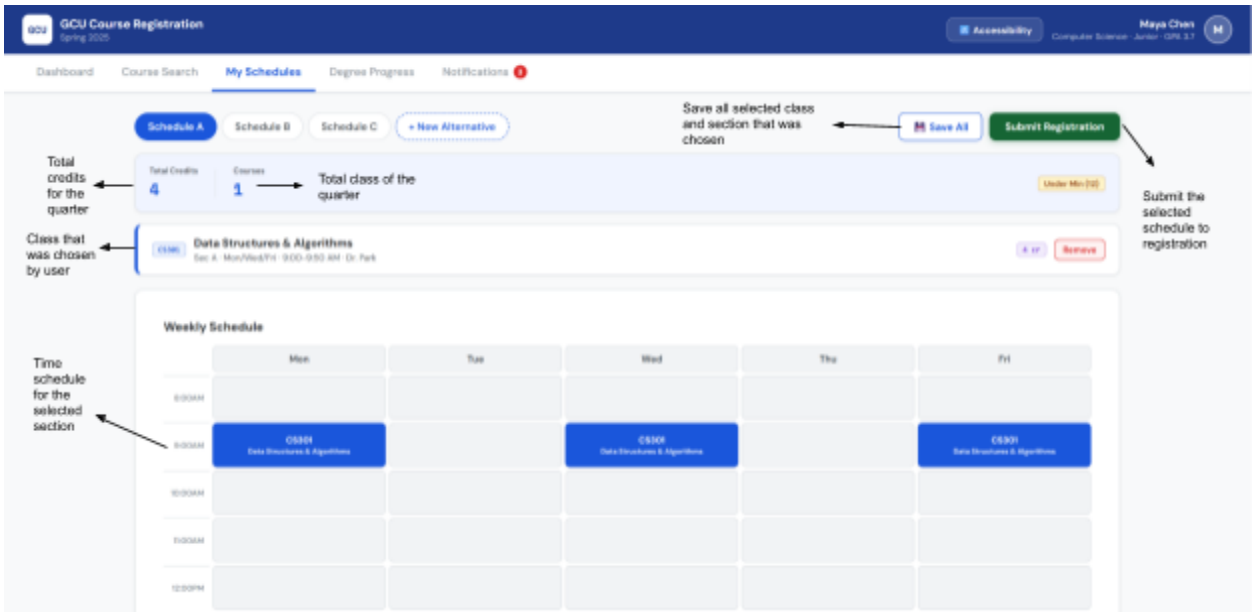
This design makes it easy for students to keep on track towards accomplishing their academic goals through easily navigable, accessible, and information-rich interfaces.

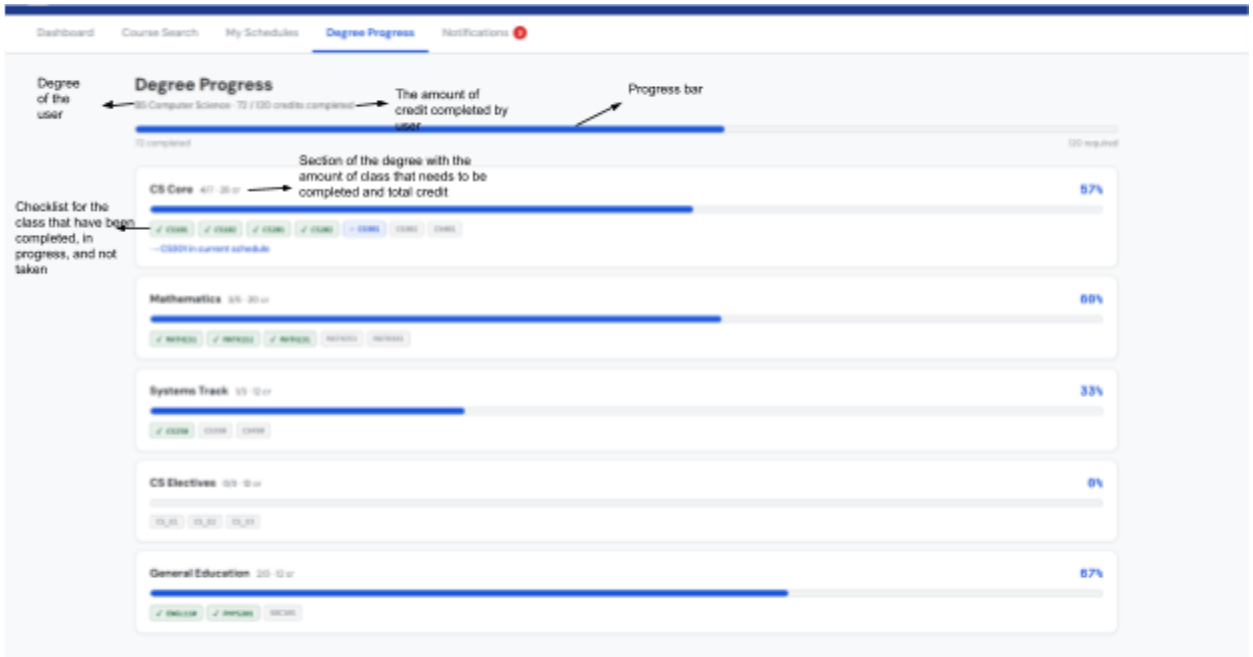
Annotated Wireframes

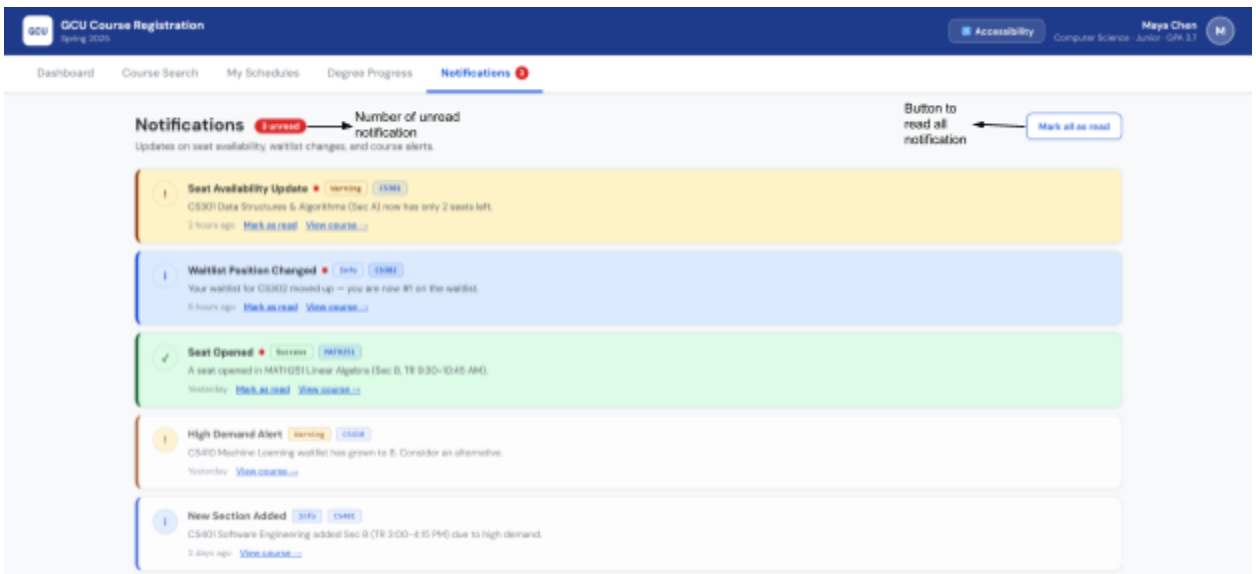

Dashboard
Course Search
My Schedules
Degree Progress
Notifications
Welcome back, Maya
Spring 2025 registration is open. Here's what we recommend.
Numbers of credit that is taken this quarter
Shows if there's any schedule conflicts
Show the alternative schedules that the users have
The system shows credit than the user have taken
CREDITS COMPLETED
72
CREDITS THIS TERM
0
SCHEDULE CONFLICTS
0
ALTERNATIVES
2
Recommended for Your Degree
Data Structures & Algorithms
Computer Organization
Linear Algebra
Human–Computer Interaction
The system recommend classes based on your degree
Courses You Need
Number of credits per course
Data Structures & Algorithms
Computer Organization
Indication of class reaching full capacity
Software Engineering
Indication of missing prerequisite


GCU Course Registration
Spring 2025
Accessibility
Maya Chen
Dashboard
Course Search
My Schedules
Degree Progress
Notifications
A search box for the classes
Search by course ID, name, or category...
Adding to:
Schedule A
Schedule B
Schedule C
Options to add classes on alternative schedule
Classes recommended based on your degree
Data Structures & Algorithms
Computer Organization
Software Engineering
Linear Algebra
Probability & Statistics
Operating Systems
Computer Networks
Machine Learning


GCU Course Registration
Spring 2025
Accessibility
Maya Chen
Computer Science · Junior · GPA 3.7
Dashboard
Course Search
My Schedules
Degree Progress
Notifications 0
Search by course ID, name, or category...
Adding to:
Schedule A
Schedule B
Schedule C
Data Structures & Algorithms
Prerequisites:
Prerequisites checklist to show that the user have or have not completed it
Fulfills: CS Core
Sec A — MWF 9:00–9:50 AM (Dr. Park) ✓
Sec B — TR 11:00 AM–12:15 PM (Dr. Liu) +
Sections for the class
Computer Organization
Software Engineering
Linear Algebra
Probability & Statistics
Operating Systems
Computer Networks
Machine Learning


GCU Course Registration
Spring 2025
Accessibility
Maya Chen
Computer Science · Junior · GPA 3.7
Dashboard
Course Search
My Schedules
Degree Progress
Notifications 0
Schedule A
Schedule B
Schedule C
+ New Alternative
Save all selected class and section that was chosen
Save All
Submit Registration
Submit the selected schedule to registration
Total credits for the quarter
Total Credits
4
Courses
1
Total class of the quarter
Under Min (12)
Class that was chosen by user
Data Structures & Algorithms
Sec A · Mon/Wed/Fri · 9:00–9:50 AM · Dr. Park
Remove
Weekly Schedule
Time schedule for the selected section
Mon
Tue
Wed
Thu
Fri
9:00AM
CS301
Data Structures & Algorithms
CS301
Data Structures & Algorithms
CS301
Data Structures & Algorithms
10:00AM
11:00AM
12:00PM





Functionality Summary

- **Course search and filtering.** The search bar filters across course ID, course name, and category. Category filter chips (CS Core, Mathematics, Elective, Gen Ed) can be combined with keyword searches to narrow results. Expanding a course card does not clear the active filter or search query.
- **Prerequisite enforcement.** Courses with unmet prerequisites are visually dimmed and their section buttons are disabled. The prerequisite status of each required course is shown inline; green tags indicate completed prerequisites and red tags indicate missing ones. Students cannot add an ineligible course to any schedule.

- **Enrollment fill indicators.** Each course card displays a mini progress bar reflecting current enrollment relative to section capacity. Fill bars shift from blue to yellow at 80% capacity and to red at 100%. Waitlist position is shown in place of open seat count when a section is full.
- **Alternative schedule planning.** Students can maintain multiple named schedules (Schedule A, Schedule B, etc.) and switch between them using the tab row in My Schedules. Adding a new alternative creates an empty schedule and sets it as active. Course additions always apply to the current active schedule.
- **Degree progress tracking.** The Degree Progress tab displays completion percentage per requirement group based on the student's completed course history. Courses currently added to the active schedule are indicated with an arrow tag, distinguishing in-progress enrollment from completed coursework. Courses that are gray indicate a need to complete.
- **Notification panel.** Each notification displays a type badge (Success, Info, Warning, Action Required), a course tag, timestamp, and read status. Unread notifications are visually highlighted. Users can mark individual notifications as read, mark all as read, dismiss individual notifications, or navigate directly to a related course from within the notification.
- **Streamlined registration submission.** Introduces a Submit Registration button within the My Schedules tab, eliminating the need to navigate to a separate submission page or system. The button is contextually placed alongside the student's build schedule, so submission happens in the same view where courses are reviewed. The button is disabled by default and becomes active when at least one course has been added with no time conflicts, preventing incomplete or conflicting submissions.
- **Increased accessibility features.** The improved interface includes several features that align with WCAG 2.1 guidelines. These features can be accessed via the "Accessibility" button at the top of the page, including high contrast mode, text scaling, and multi-language support (currently offering 3 language options). After users make their desired changes they can close the drop down menu by simply clicking the accessibility button again.

Related Backlog Items

Epics:

- T1I3W2-3: Student Planning and Visibility
- T1I3W2-2: Reliable and Fair Course Registration

User Stories:

- T1I3W2-10: Prerequisites Checked Automatically
- T1I3W2-9: Multiple Schedule Options

- T1I3W2-34: Academic Progress Transparency
- T1I3W2-41: Plan Courses Ahead
- T1I3W2-39: Unmet Prerequisite Identification

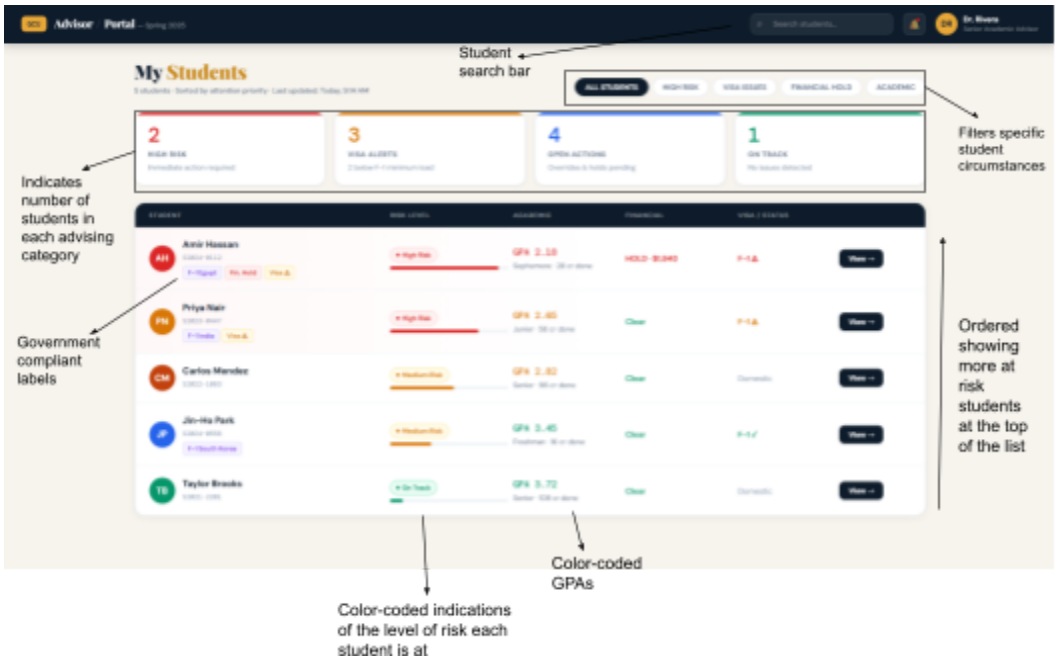
Academic Planning Dashboard

GSU Academic Advising Dashboard

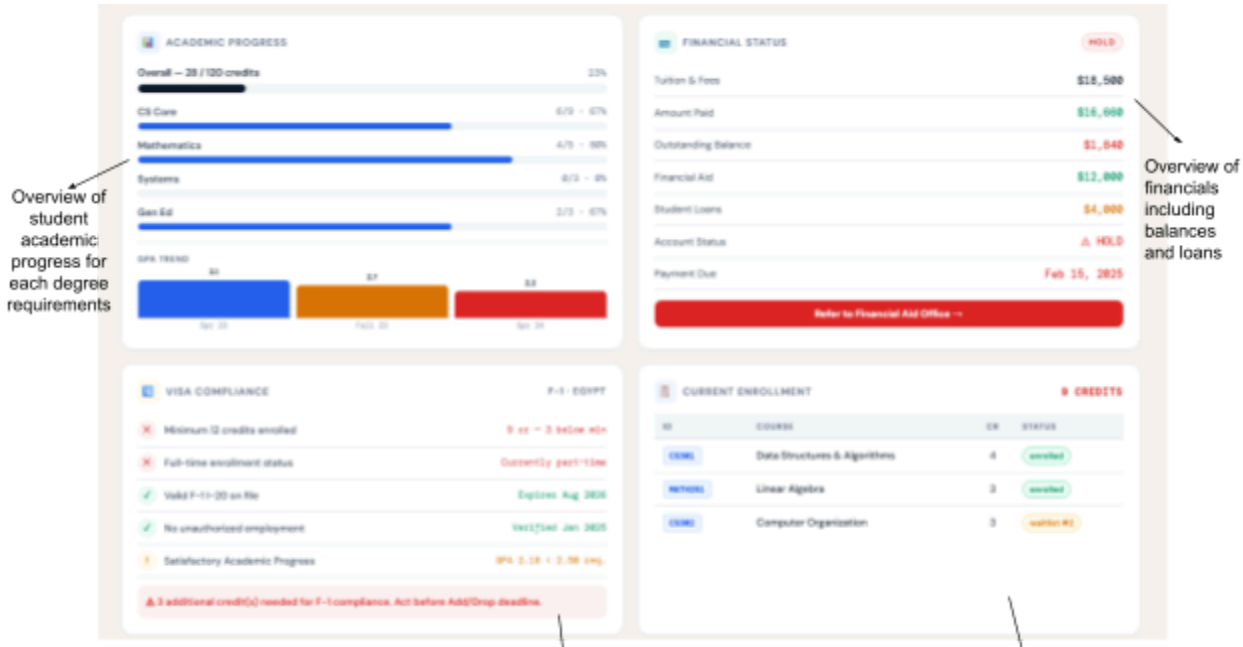
The Academic Advising Dashboard is designed to help advisors quickly identify students who may need support and monitor their academic progress in one place. The main interface highlights key student information such as risk level, GPA, hold flags, and visa compliances. Advisors can easily search for students and filter by alerts. When advisors click on individual students, they view a detailed overview of student profiles that bring together academic, financial, and enrollment information in one place.

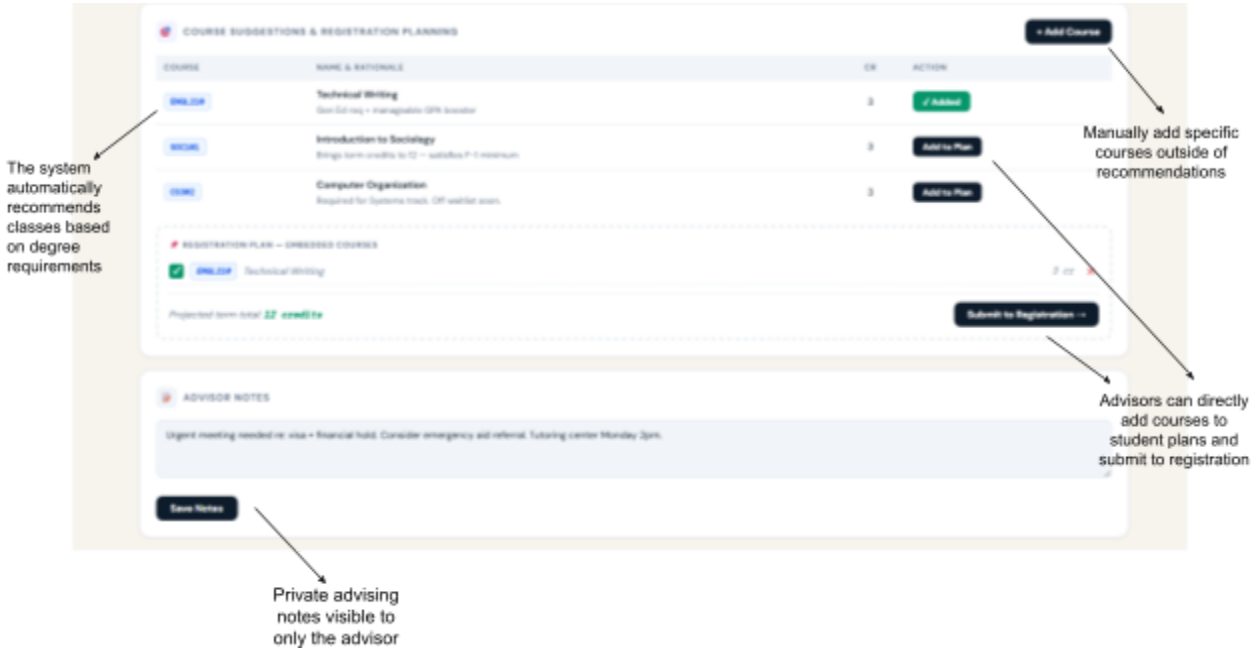
By consolidating this information into a single dashboard, the interface helps advisors spend less time navigating multiple systems and more time supporting students, which was recognized as a challenge for this stakeholder. The dashboard also helps advisors proactively identify students who may be falling behind, allowing them to recommend courses, schedule advising meetings, and approve necessary overrides. This improves coordination between advising, financial aid, and compliance offices while helping students stay on track towards graduation.

Annotated Wireframes



Advisor Portal
← BACK TO ALL STUDENTS
AH
Amir Hassan
2.18
28
9
23%
HOLD
ALERTS & ACTION ITEMS
Total overview of student including academics and government compliance
Ways to contact student
Streamlined override request button
A list of alerts and actions ranked by importance


ACADEMIC PROGRESS
Overall — 28 / 120 credits
23%
CS Core
6/9 · 67%
Mathematics
4/5 · 80%
Systems
0/3 · 0%
Gen Ed
2/3 · 67%
GPA TREND
FINANCIAL STATUS
HOLD
Tuition & Fees
$18,500
Amount Paid
$16,000
Outstanding Balance
$1,040
Financial Aid
$12,000
Student Loans
$4,000
Account Status
⚠ HOLD
Payment Due
Feb 15, 2025
Refer to Financial Aid Office →
VISA COMPLIANCE
F-1 · EGYPT
Minimum 12 credits enrolled
0 cr — 3 below min
Full-time enrollment status
Currently part-time
Valid F-1/I-20 on file
Expires Aug 2026
No unauthorized employment
Verified Jan 2025
Satisfactory Academic Progress
GPA 2.18 < 2.50 req.
⚠ 3 additional credit(s) needed for F-1 compliance. Act before Add/Drop deadline.
CURRENT ENROLLMENT
9 CREDITS
ID
COURSE
CR
STATUS
CS301
Data Structures & Algorithms
4
enrolled
MATH201
Linear Algebra
3
enrolled
CS302
Computer Organization
3
waitlist #2
Overview of student academic progress for each degree requirements
Overview of financials including balances and loans
Student's visa compliance statuses and flags
Shows students class enrollments and statuses



Functionality Summary

- **Student search and filtering.** Advisors can search for students using the search bar. Filter chips allow advisors to narrow the list by issue type, such as high-risk students, visa compliance concerns, financial holds, or other academic alerts. These filters can be combined with keyword searches to quickly locate specific students who may require attention.
- **Risk prioritization dashboard.** The interface displays summary cards highlighting the number of high-risk students, visa alerts, open actions, and students currently on track. Each student row includes a visual risk indicator and a progress meter that reflects the student's overall risk score (high risk, medium risk, on track). This allows advisors to immediately identify which students require urgent intervention and to prioritize outreaches.
- **Student profile overview.** Selecting a student opens a detailed profile that consolidates essential academic and administrative information into a single interface. Advisors can view student identifiers, program information, GPA, credit progress towards degree completion, enrollment statuses, and financial standings. Key alerts and compliance flags are displayed prominently at the top of the profile to ensure advisors can quickly understand the student's current situation.
- **Academic progress tracking.** The academic progress panel displays degree completion progress across requirement categories such as core courses and general education. Progress bars help visually represent the proportion of completed courses relative to program requirements. Advisors can also view GPA trends over previous semesters through a simple visual chart, allowing them to identify potential academic declines early.

- **Compliance monitoring tools.** The interface includes dedicated checks for compliance requirements such as international student visa status and enrollment minimums. Visual indicators highlight whether a student meets requirements. This allows advisors to quickly identify compliance risks that may affect a student's legal or financial standing.
- **Financial status visibility.** Advisors can view a summary of tuition charges, payments, financial aid, loans, and any outstanding balances. Financial holds are clearly flagged to indicate when a student may be restricted from registering for courses. By displaying financial information alongside academic records, the system helps advisors understand barriers that may prevent students from completing registration.
- **Advising action tools.** Advisors can take immediate action directly from the student profile through built-in controls such as emailing, scheduling meetings, requesting enrollment overrides, and saving advising notes. These tools reduce the need to switch between systems and allow advisors to document interactions and interventions within the same interface used for monitoring student progress.

Related Backlog Items

Epics:
- T1I3W2-3: Student Planning and Visibility
- T1I3W2-4: Administrative Control and Oversight

User Stories:
- T1I3W2-11: Granting Overrides
- T1I3W2-22: Automated Enrollment Monitoring
- T1I3W2-24: Financial Status Verification
- T1I3W2-25: Financial Implication Visibility
- T1I3W2-35: Student Academic Progress Visibility
- T1I3W2-36: High-Risk Student Identification
- T1I3W2-45: Student Course Planning

Department Administrator Dashboard

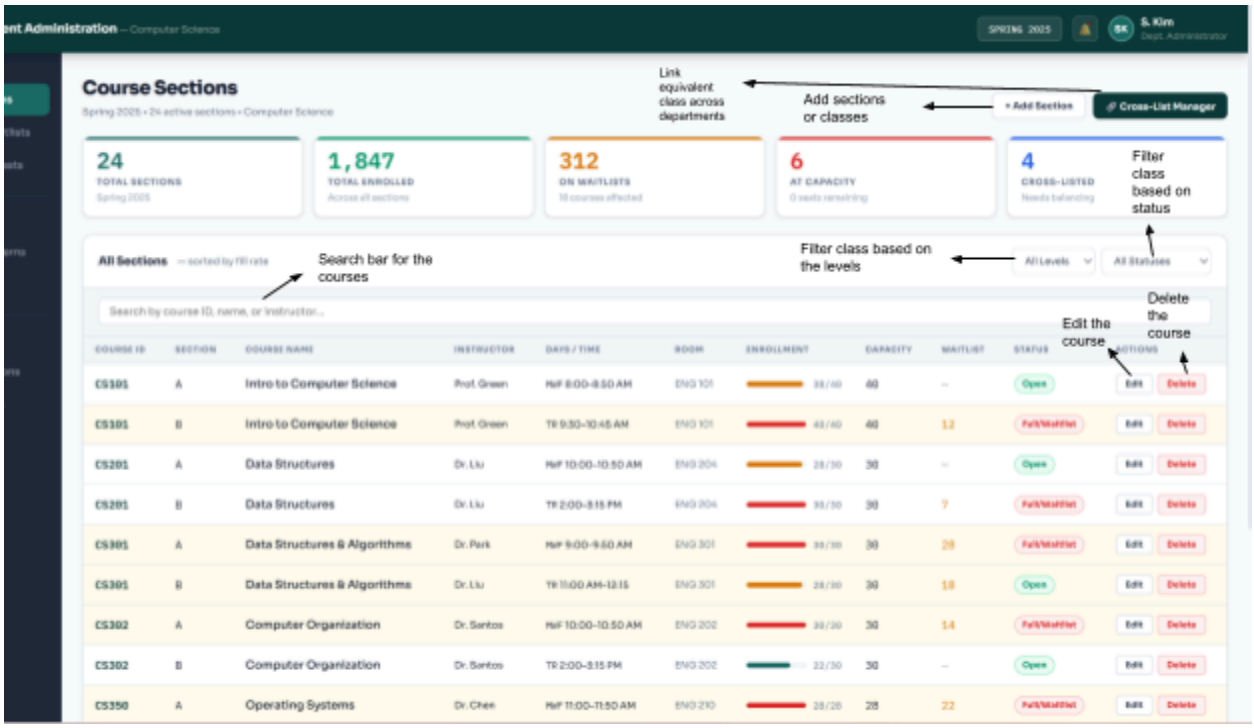
GSU Department Administration Dashboard

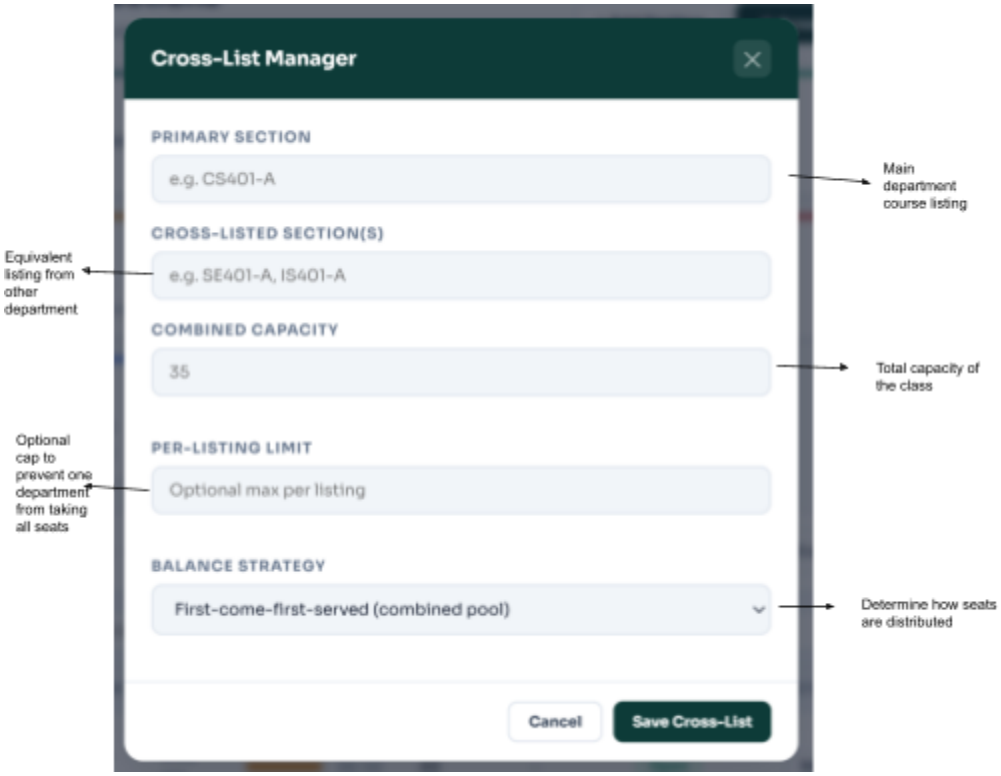
The Department Administrator Dashboard is designed to help department administrators efficiently manage course sections and respond to enrollment demand in one place. The main interface highlights key operational information such as section capacities, waitlist pressure, override requests, and registration activity, allowing administrators to make quick decisions without navigating between multiple systems.

The design consolidates course management, capacity adjustments, demand reporting, and audit logging into a single unified interface, directly addressing the administrative challenges identified in the stakeholder interviews.

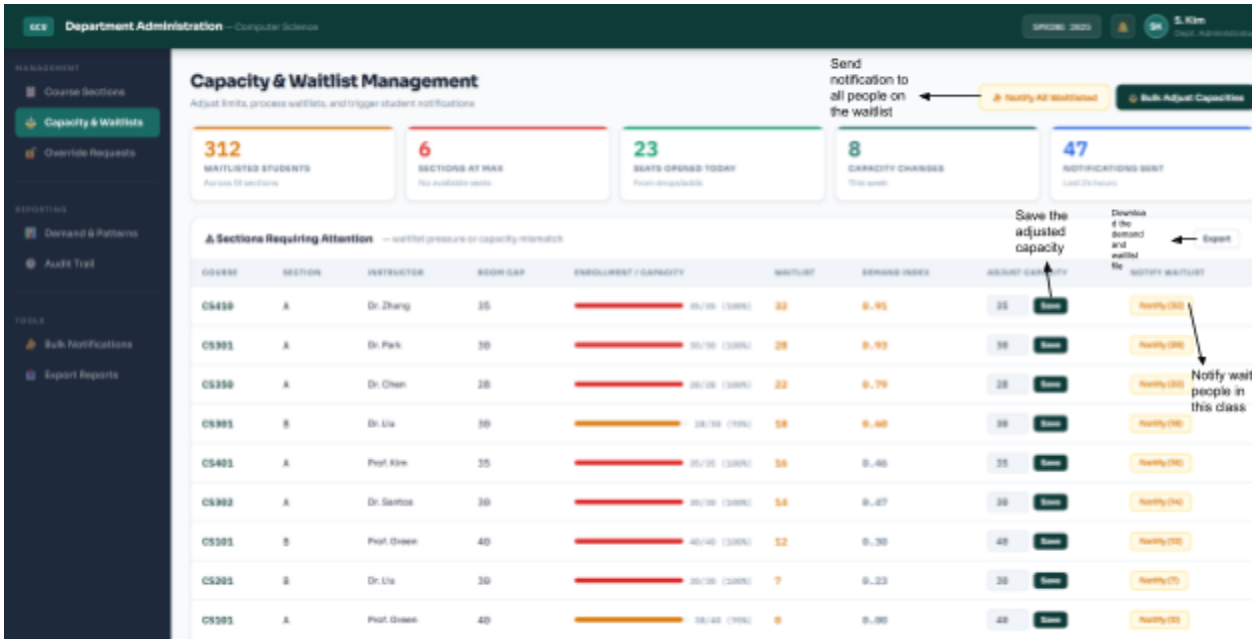
The dashboard also provides administrators with the reporting and oversight tools needed to plan ahead and maintain accountability. The Demand & Patterns panel surfaces waitlist pressure data and enrollment trends to inform future course planning, while the Audit Trail ensures every capacity change, override decision, and enrollment action is permanently logged with a timestamp and user identity. Together these features give department administrators the visibility and control needed to manage a complex registration system with confidence.

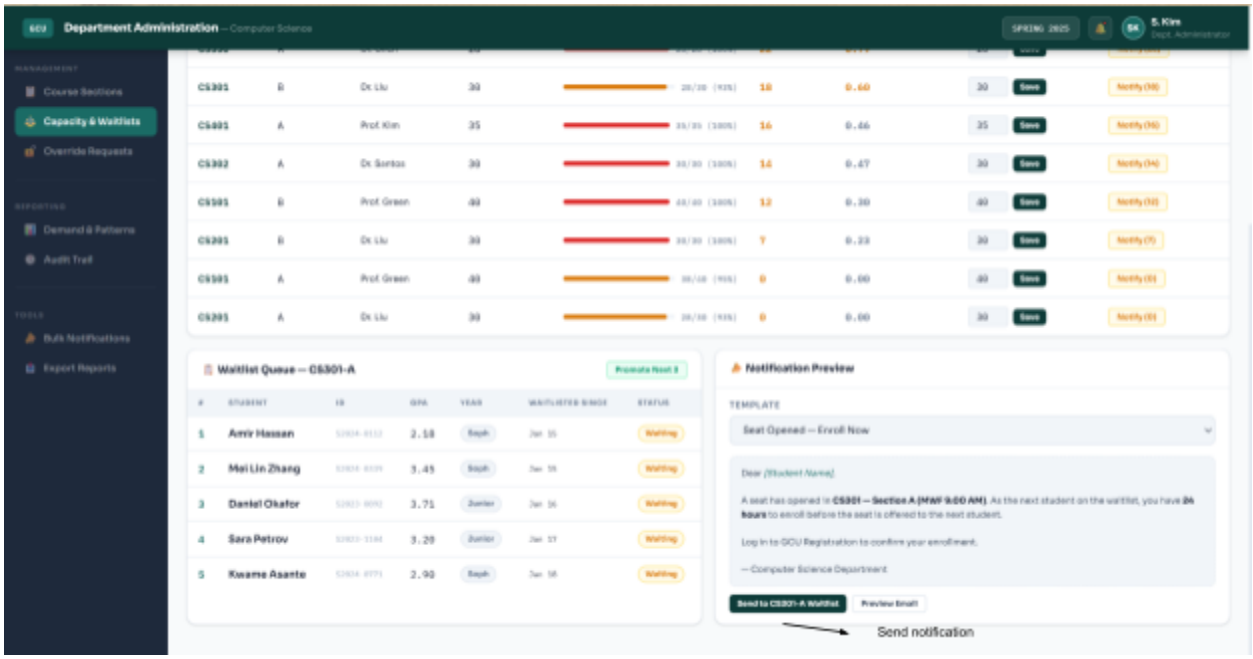
Annotated Wireframes

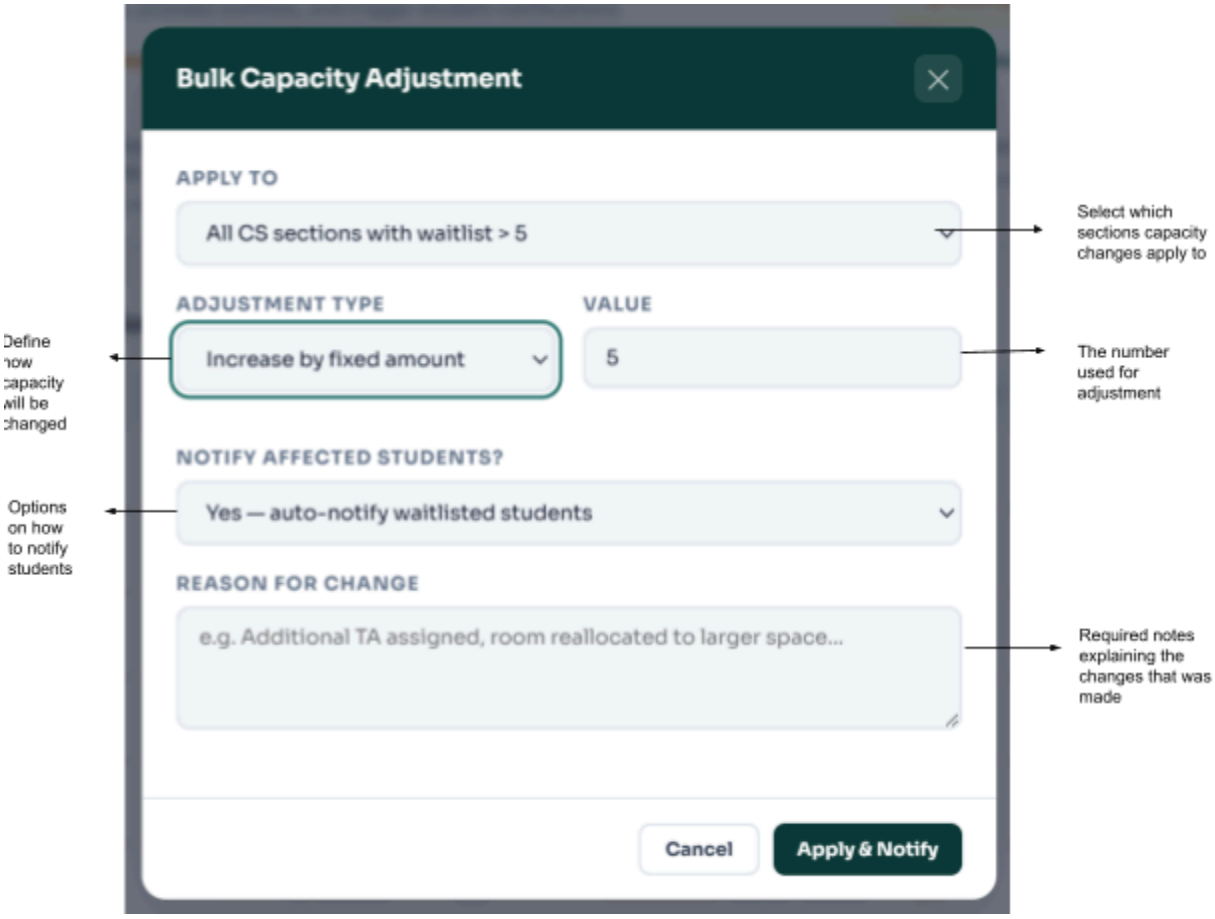


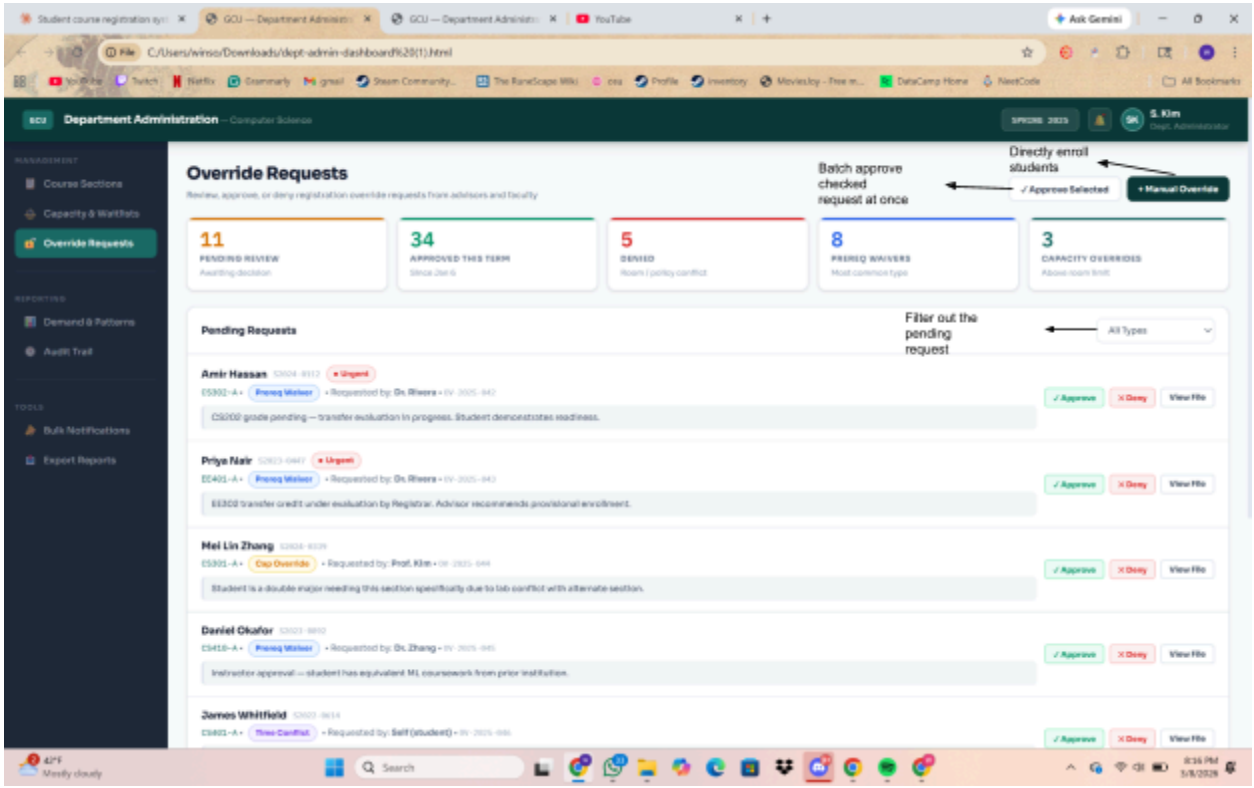

Cross-List Manager
PRIMARY SECTION
e.g. CS401-A
CROSS-LISTED SECTION(S)
e.g. SE401-A, IS401-A
COMBINED CAPACITY
35
PER-LISTING LIMIT
Optional max per listing
BALANCE STRATEGY
First-come-first-served (combined pool)
Cancel
Save Cross-List
Main department course listing
Equivalent listing from other department
Total capacity of the class
Optional cap to prevent one department from taking all seats
Determine how seats are distributed

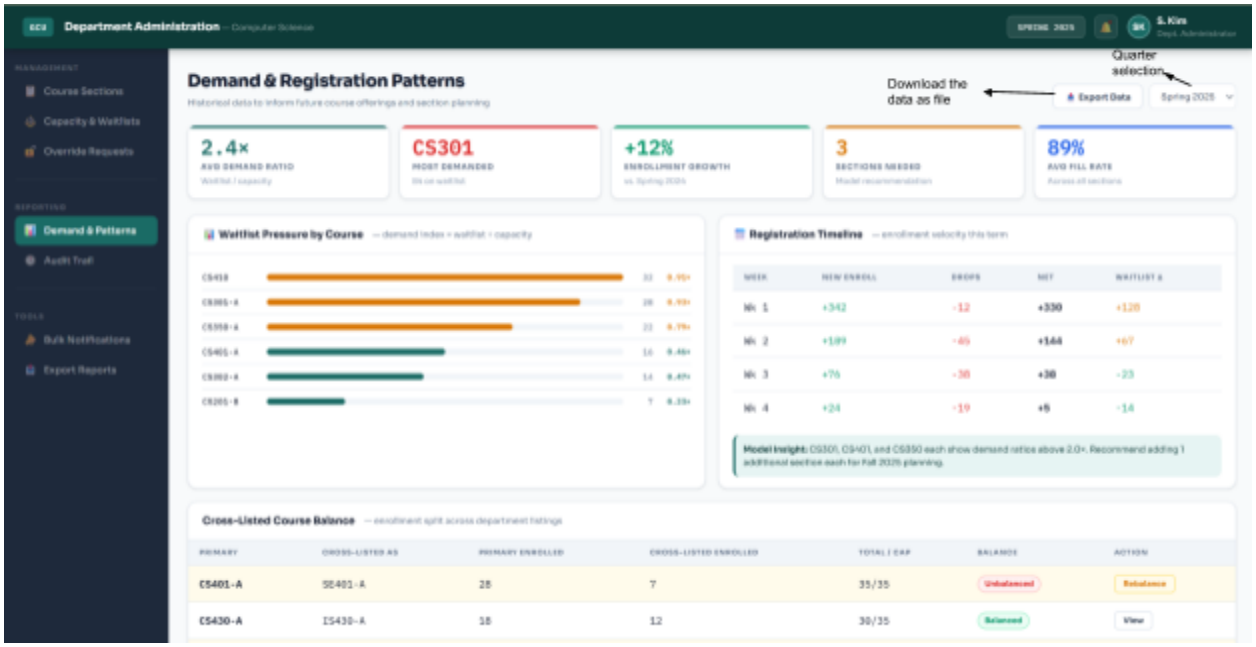
Edit Section CS101-A
COURSE ID
CS101
SECTION
A
COURSE NAME
Intro to Computer Science
INSTRUCTOR
Prof. Green
MEETING DAYS
MWF
TIME
8:00–8:50 AM
ROOM
ENG 101
CAPACITY
40
STATUS
Open
Changes will be saved to the section record and reflected in the table immediately.
Cancel
Save Changes
Section of the course
Name of the course
The time of the week the class meets
Total capacity of the class
Status of the class
The days of the week the class meets
Room location of the class


Department Administration — Computer Science
S. Kim
Course Sections
Capacity & Waitlists
Override Requests
Demand & Patterns
Audit Trail
Bulk Notifications
Export Reports
Capacity & Waitlist Management
Adjust limits, process waitlists, and trigger student notifications
Send notification to all people on the waitlist
Bulk Adjust Capacities
312
WAITLISTED STUDENTS
6
SECTIONS AT MAX
23
SEATS OPENED TODAY
8
CAPACITY CHANGES
47
NOTIFICATIONS SENT
Sections Requiring Attention
Save the adjusted capacity
Export
Notify wait people in this class


Department Administration — Computer Science
SPRING 2025
S. Kim
Course Sections
Capacity & Waitlists
Override Requests
Demand & Patterns
Audit Trail
Bulk Notifications
Export Reports
Waitlist Queue — CS301-A
Promote Next 3
Notification Preview
TEMPLATE
Seat Opened — Enroll Now
Send to CS301-A Waitlist
Preview Email
Send notification


Bulk Capacity Adjustment
APPLY TO
All CS sections with waitlist > 5
Select which sections capacity changes apply to
ADJUSTMENT TYPE
VALUE
Define how capacity will be changed
Increase by fixed amount
5
The number used for adjustment
NOTIFY AFFECTED STUDENTS?
Options on how to notify students
Yes — auto-notify waitlisted students
REASON FOR CHANGE
e.g. Additional TA assigned, room reallocated to larger space...
Required notes explaining the changes that was made
Cancel
Apply & Notify


Department Administration
Override Requests
Batch approve checked request at once
Directly enroll students
Filter out the pending request
11
34
5
8
3
Pending Requests
Amir Hassan
Priya Nair
Mei Lin Zhang
Daniel Okafor
James Whitfield



Functionality Summary

- **Course section management.** The administrator can view all active course sections in a searchable, filterable table. Sections can be filtered by level and status. Each row displays enrollment fill bars, waitlist counts, and section status. Administrators can edit section details such as instructor, room, meeting days, time, capacity, and status directly through an edit modal that updates the table immediately upon saving. Sections can also be deleted with a confirmation dialog that requires the administrator to type the section ID to prevent accidental removal.
- **Cross-listed cross management.** The Cross-List Manager allows the administrator to link a course section to equivalent listings across departments, set a shared combined capacity, apply optional per-listing limits, and choose a balancing strategy to automatically distribute enrollment between department listings.
- **Capacity and waitlist management.** The interface highlights sections at or near capacity and displays live enrollment bars and demand index scores. Administrators can adjust section capacities inline and save changes directly from the table. Waitlisted students can be notified individually per section or all at once using the Notify All Waitlisted button. The waitlist queue shows the ordered list of waiting students and allows the administrator to promote multiple students at once.
- **Bulk capacity management.** The Bulk Adjust tool allows the administrator to apply a single capacity change across multiple sections simultaneously, with options to define the adjustment type, set the value, and automatically notify affected waitlisted students. A required reason field ensures all bulk changes are documented.

- **Override request processing.** The override queue displays all pending registration exception requests submitted by advisors, instructors, or students. Each request shows the student, course, override type, requester, and justification. Administrators can approve or deny individual requests, with all decisions permanently logged in the audit trail. A Manual Override option allows administrators to directly enroll a student without a formal request.
- **Demand and registration pattern reporting.** The Demand & Patterns panel provides waitlist pressure data by course, showing demand index ratios to identify which sections have the highest unmet demand. The registration timeline tracks weekly enrollment and drop trends. Cross-listed course balance is displayed with flagged unbalanced sections and a one-click rebalance option. A model insight recommendation suggests which courses need additional sections for future planning. Data can be exported as a file for use outside the system.
- **Audit trail.** A searchable and filterable log records every administrative action taken in the system, including capacity changes, override decisions, enrollments, drops, and notifications. Each entry includes the action taken, detail, the user responsible, and a timestamp, providing a complete accountability record for the department.

Related Backlog Items

Epics:
- T1I3W2 : Administrative Control and Oversight

User Stories:
- T1I3W2-11: Grants Override
- T1I3W2-23: Analytical Dashboard
- T1I3W2-21: Adjust Course Capacities
- T1I3W2-20: Audit Log Visibility

Data Flow Diagram

The Data Flow Diagram (DFD) for the GSU Course Registration System illustrates how information moves between users, system processes, and data stores to support secure, fair, and efficient course registration. This diagram focuses on the logical flow of data needed to authenticate users, search and enroll in courses, manage waitlists and overrides, and track student progress and reporting.

External Entities

Student (external entity)
Purpose: initiates login, searches courses, submits enrollment & waitlist requests.

Key data exchanged: studentID, NetID, sessionToken, requestedCourseID, timestamp (ms).
Format/frequency: JSON over HTTPS; interactive (user-driven) requests.
Validation/security: input sanitized on client; server verifies MFA + role before processing.
Maps to: Reliable and Fair Course Registration epic; AC: timestamp precision, MFA.

Academic Advisor (external entity)
Purpose: views advisee progress, approves overrides, receives waitlist/alert notifications.
Key data exchanged: advisorID, adviseeList, overrideRequestID, approvalDecision.
Format/frequency: JSON; event-driven notifications + periodic dashboard polling.
Security: role-based access; advisors see only assigned students.
Maps to: Administrative Control & Oversight epic; AC: advisor override logging, limited view.

Faculty / Dept Admin (external entity)
Purpose: view enrollment, adjust capacities (manual override), confirm attendance for reports.
Key data exchanged: courseID, newCapacity, confirmationEntries.
Format/frequency: HTTPS form POST; low-frequency changes.
Security: scoped role access; capacity changes require justification & audit entry.
Maps to: Department admin capacity user story; AC: capacity change audit.

IT / External Services (external entity)
Purpose: authentication provider, notification/email service, SIS integration.
Key data exchanged: auth tokens, notification payloads, student records sync.
Format/frequency: secure API calls (OAuth/JWT), batch sync nightly, realtime for auth.
Security / Compliance: encrypted at rest/in transit; audit all syncs; FERPA controls.
Maps to: system integrations and security epics; AC: encryption, audit logs.

Processes

P1 — Authenticate User
Checks the user's login information and confirms their identity.
Assigns access based on their role (student, advisor, or administrator).

P2 — Search & Register Course
Allows students to search for courses and submit registration requests.
Verifies prerequisites, schedule conflicts, and seat availability before confirming enrollment.

P3 — Manage Waitlist & Overrides
Processes waitlist updates and advisor override decisions.
Updates enrollment status and sends notifications when changes occur.

P4 — Track Progress & Reports
Uses enrollment data to generate student progress dashboards and system reports.
Provides alerts and summaries for advisors, faculty, and administrators.

Data Stores

D1 — Course Catalog
Stores course details such as course name, schedule, capacity, and prerequisites.
Used when students search for and view course information.

D2 — Enrollment & Waitlist
Stores current enrollments and waitlist positions for each course.
Updates when students register, drop courses, or when overrides are approved.

D3 — Student Records
Stores student information such as academic status, current credits, and program details.
Used to verify eligibility and check requirements before registration.

D4 — Degree Audit & Logs
Stores student progress toward degree completion and keeps a record of registration activity.
Used to generate progress dashboards and maintain compliance records.

Key Data Flows

Login Request
Student → Authenticate User
Student enters login information to access the system.

Verify Identity
Authenticate User → Student Records
System checks student account details and confirms role and permissions.

Search Request
Student → Search & Register Course
Student searches for available courses.

Fetch Course Information
Search & Register Course → Course Catalog
System retrieves course details such as schedule, capacity, and prerequisites.

Enrollment Request
Student → Search & Register Course
Student selects a course and submits a registration request.

Store Enrollment / Waitlist
Search & Register Course → Enrollment & Waitlist
System records the student's enrollment or adds them to the waitlist.

Registration Confirmation
Search & Register Course → Student
Student receives confirmation or error message after attempting registration.

Override Approval
Advisor → Manage Waitlist & Overrides
Advisor reviews and approves or denies special registration requests.

Update Enrollment / Waitlist
Manage Waitlist & Overrides → Enrollment & Waitlist
System updates enrollment status or waitlist order after advisor action.

Log Activity
Track Progress & Reports → Degree Audit & Logs
System records registration activity for reporting and compliance.

Progress Reports
Track Progress & Reports → Advisor / Faculty / IT Dept.
System provides dashboards and reports showing student progress and registration trends.

Data Flow Diagram

GSU Course Registration System Data Flow Diagram

Epics:

- T1I3W2: Reliable and Fair Course Registration
- T1I3W2-3 : Student Planning and Visibility
- T1I3W2-4: Administrative Control and Oversight
- T1I3W2-49: Security, Privacy and Compliance

User Stories:

- T1I3W2-39: Clear Prerequisite Blocking
- T1I3W2-34: View Degree Progress
- T1I3W2-11: Advisor Grant Registration Override
- T1I3W2-50: Multi-Factor Authentication